LIN YU XIANG

Taichung, Taiwan

PROFESSIONAL SUMMARY

Electrical Engineering graduate student with hands-on experience in automation system integration, industrial computer vision, C# SDK development, and network troubleshooting. Experienced in hardware-software integration, technical problem-solving, and cross-functional collaboration.

TECHNICAL SKILLS

- Programming & Development: Python, C, C#, SDK Integration, Web Development
- Computer Vision & Automation: PyTorch, YOLOv5, Automated Optical Inspection (AOI)
- Equipment & Networking: Cisco Switches, Network Troubleshooting

EDUCATION

National Chung Cheng University , M.S. in Electrical Engineering — *2025 - Present*

- Expected Graduation: Aug 2027
- Research Area: Signal Processing, Image Processing, and Intelligent Computing

National Yunlin University of Science and Technology, B.S. in Electrical Engineering — *2021 - 2025*

- Automation Engineering Program | 2024

WORK EXPERIENCE

Part-Time Research Assistant (Contract) — *Mar 2026 – Present*

National Taiwan University Hospital, Yunlin Branch

- Developed experimental interfaces and eye-tracking data acquisition functions using C# and the Ganzin SDK.
- Assisted in eye-tracking experiment design, eye-tracking data collection, and equipment procurement.

Student Dormitory Network Management Team — *Sep 2021 – Jun 2025*

National Yunlin University of Science and Technology

- Maintained network equipment and diagnosed connectivity issues to support network operations.
- Assisted with website maintenance, system troubleshooting, and technical support.
- Held multiple team roles, including Team Leader, coordinating members and supporting technical training.

ACADEMIC PROJECTS

Automated Production Line Integration and AI-Based Optical Defect Detection

- Participated in automated assembly line development involving multiple robotic arms, workflow planning, and fixture configuration.
- Developed a YOLOv5-based defect detection model using PyTorch to identify missing components, assembly defects, and surface scratches.
- Helped streamline product changeovers through fixture adjustments and central control program switching.

RESEARCH EXPERIENCE

Multimodal Emotion Recognition Using EEG and Eye-Tracking Signals

- Investigating multimodal emotion recognition using EEG and eye-tracking features from the SEED-IV dataset.
- Exploring brain-region analysis and cross-attention mechanisms for multimodal feature fusion and emotion classification.

ADDITIONAL EXPERIENCE

Laboratory Coordinator — *Mar 2026 – Present*

- Equipment management, procurement planning, and laboratory coordination.

Teaching Assistant — *Feb 2026 – Present*

- Laboratory instruction, course materials, and student support.

Part-Time Administrative Assistant — *Jan 2024 – Jun 2025*

- Documentation and departmental coordination.

HONORS & AWARDS

- **Honorable Mention**, NYUST Electrical Engineering Project Competition *2024*
- **Honorable Mention**, 3rd The Chenbro-Syntec Cup Smart Robotics Industry-Academia Training Camp *2024*
- **Silver Award**, 16th Artificial Intelligence and Data-Driven Chemical Intellectual Property Competition *2023*